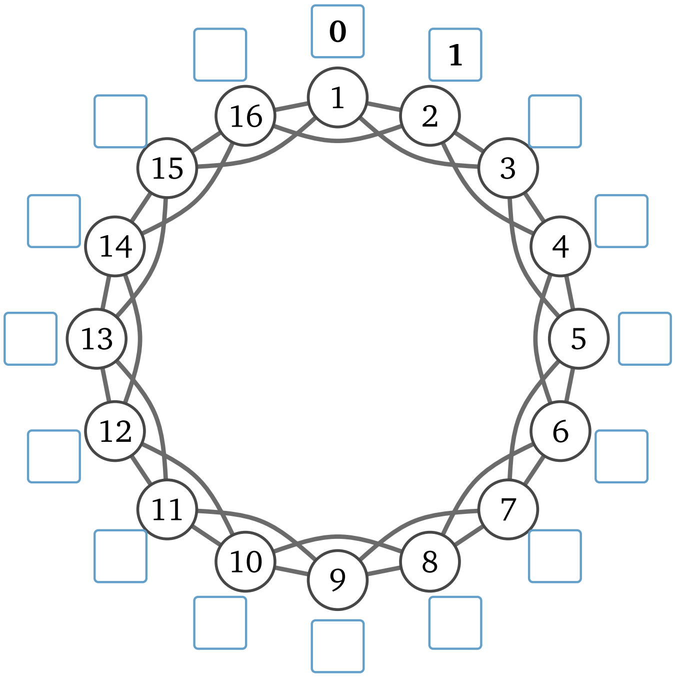
Part 1 — Ringville

Sixteen people live in Ringville and they sit in one circle. Everybody is friends with the two on their left and the two on their right: four friends each, thirty-two friendships in the town.



Question 1(a): Person 1 has a letter for person 9. Trace the chain on the drawing. How many handshakes does it take?

Question 1(b): Now everybody. In each person's box on the drawing, write how many handshakes it takes to reach them from person 1. Two are written in already, and you may find you only have to work out half of the rest.

Question 1(c): Add up every number in the boxes. What is the total?

Share that total out among the fifteen other people. That is the town's **average**.

And what is the biggest number in any box — the town's **worst case**?

Question 1(d): A newspaper wants one number for a headline: "in Ringville, two people are X handshakes apart."

The post office wants one number for a promise it prints on the postbox: "every letter arrives in X handshakes or fewer."

You have two numbers, the average and the worst case. Which one goes in the headline, and which one goes on the postbox?

What goes wrong if you swap them?

Question 2: Ringville grows. A thousand people, still one circle, still four friends each.

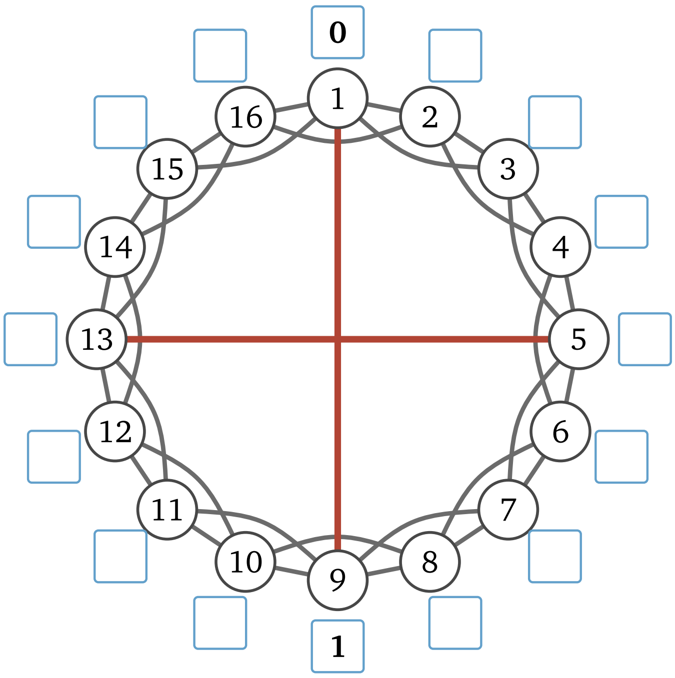
How many handshakes does it take to reach the person sitting directly opposite you?

And if the circle held all eight billion people on Earth?

Could a world wired like Ringville pass a letter from Omaha to Boston in six handshakes? Why?

Part 2 — Two shortcuts

Two of the sixteen went to college and came home with a friend from the far side of town: person 1 is now also friends with person 9, and person 5 with person 13 — thirty-four friendships, not thirty-two.



Question 3: Write the handshake counts into the boxes again, this time for the town with the two new friendships. Two are written in already.

Then work out the same three numbers as in 1(c): the new total, the new average, and the new worst case.

Question 4: Thirty-two friendships became thirty-four. Work out both changes as percentages: by how much did the number of friendships change, and by how much did the average change?

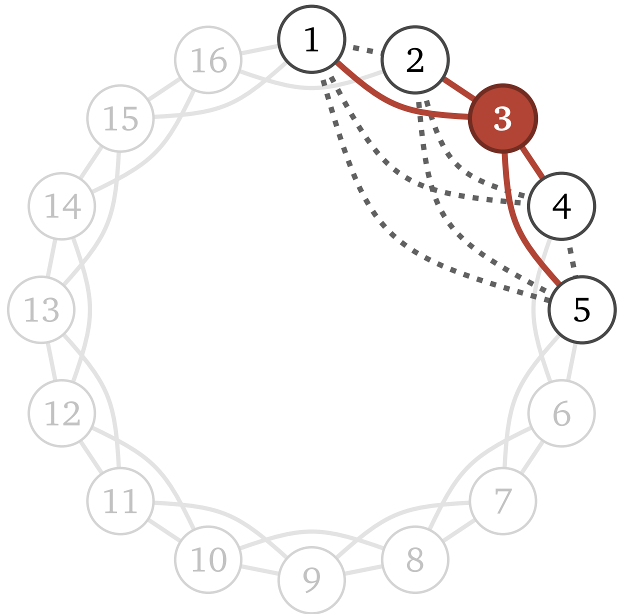
Say, in your own words, what those two new friendships did that thirty-two ordinary ones could not.

Part 3 — Do your friends know each other?

Ringville again, as it was in Part 1, before the two shortcuts. Take somebody ordinary: **person 3**, in red, whose four friends are persons 1, 2, 4 and 5.

Four people (1, 2, 4, 5) can be paired up six ways: $(1,2)$, $(1,4)$, $(1,5)$, $(2,4)$, $(2,5)$, $(4,5)$. Those six pairs are the dashed lines.

Darken the dashed lines that are real friendships.



Question 5(a): Ringville grows to ten thousand people, still four friends each. How many of person 3's six pairs are friends now?

Question 5(b): A different town: the same sixteen people and the same thirty-two friendships, but this time each friendship is drawn between two people uniformly at random.

Pick two people. What is the chance that they are friends?

Each of person 3's six pairs is friends with that same chance. So out of six pairs, how many would you expect to be friends?

And in a random town of ten thousand, still four friends each?

Question 5(c): Take the number of person 3's six pairs that really are friendships, and divide it by six. That fraction is person 3's **local clustering**. You have now worked it out four times: Ringville, Ringville with ten thousand people, and the two random towns.

Compare the four numbers. What is it about the way friends are made in each of the two towns that decides whether local clustering survives when the town grows?

Three people who are all friends with each other make a **triangle**. Ringville, before the two shortcuts, has sixteen of them.

Question 6(a): Now go back to the drawing in Part 2, where the two shortcuts have gone in.

Do persons 1 and 9 have a friend in common? Do persons 5 and 13?

So how many triangles does the town have now?

Question 6(b): In Part 2 the two shortcuts were new friendships. The real model never adds one: it takes a friendship away and puts it back somewhere else.

Do that once: person 1 stops being friends with person 2, and becomes friends with person 9 instead. The town still has thirty-two friendships.

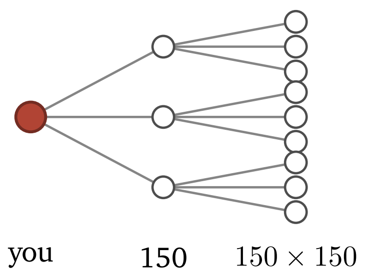
Which friends did persons 1 and 2 have in common?

So how many triangles does the town lose?

Question 6(c): What does a shortcut buy the town, and what does it cost?

Part 4 — Six handshakes?

Question 7(a): Everybody knows 150 people. Pretend for a moment that no two of your friends know each other, so every handshake reaches 150 people nobody in the chain has met yet. Round as you like.



drawn with 3 friends each, not 150

handshakes	people reached
1	150
2	22,500
3	about 3.4 million
4	
5	

Carry the table two rows further. At how many handshakes do you pass eight billion? Compare that with the guess you made at the start of class.

Question 7(b): Look at the tree again. Every branch lands on somebody new, so no person appears in it twice.

But in Part 3 you found that your friends are often friends with each other. So some branches must land on a person who is already in the tree.

In 7(a) you counted the handshakes it takes to reach eight billion people. In a real town, is the true number bigger or smaller than the one you counted? Why?

Part 5 — The lab, on your own

Do this one by yourself, with this sheet beside you. The notebook counts the same town's handshakes and triangles for you. The marked cells in it are blank and yours to fill.

Point a camera at the square, or type `go.skojaku.com/m02lab`